

1.9.6

EE25BTECH11017 - Mahesh Chollangi.

Question:

If $\mathbf{Q} = (0, 1)$ is equidistant from $\mathbf{P} = (5, -3)$ and $\mathbf{R} = (x, 6)$, find the value of x .
(10, 2023)

Solution:

Variable	Description
$\mathbf{P}$	Point $(5, -3)$
$\mathbf{Q}$	Point $(0, 1)$
$\mathbf{R}$	Point $(x, 6)$

TABLE 0: Variables Used

$$\mathbf{P} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} x \\ 6 \end{pmatrix} \quad (1)$$

Given that point $\mathbf{Q}$ is equidistant from the points $\mathbf{P}$ and $\mathbf{R}$, which is given by

$$\|\mathbf{Q} - \mathbf{P}\| = \|\mathbf{Q} - \mathbf{R}\| \quad (2)$$

Substituting values,

$$\left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ -3 \end{pmatrix} \right\| = \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} x \\ 6 \end{pmatrix} \right\| \quad (3)$$

$$\left\| \begin{pmatrix} -5 \\ 4 \end{pmatrix} \right\| = \left\| \begin{pmatrix} -x \\ -5 \end{pmatrix} \right\| \quad (4)$$

$$\sqrt{(5)^2 + (4)^2} = \sqrt{(x)^2 + (5)^2} \quad (5)$$

$$(6)$$

Simplifying,

$$(5)^2 + (4)^2 = (x)^2 + (5)^2 \quad (7)$$

$$(4)^2 = (x)^2 \quad (8)$$

$$\implies x = 4 \quad \text{and} \quad x = -4 \quad (9)$$

Hence, the required values of x are : $x = 4$ and $x = -4$

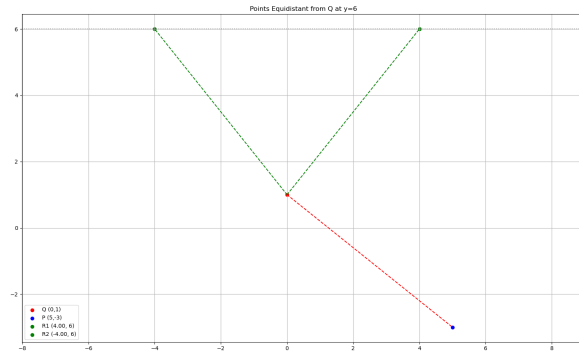


Fig. 0.1: plot